

Modbus

Quick Guide for Bonfiglioli Inverter

Bonfiglioli-Vectron ReSC

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1 Introduction

Modbus definitions for reading/writing data in a device do not fit directly to parameter access of inverters (independent of inverter manufactures). Therefore it is necessary to map the register to the parameter number.

The current inverter software supports access to 16-Bit and 32-Bit parameters via Modbus Function Code 0x03 Read Holding Registers, 0x06 Write Single Register and 0x10 Write Multiple Registers.

The Modbus Block Read function allows the user to read up to 64 16-Bit parameters in one block transfer. For the Block Read function a different access method is required which does not conflict with the current method of accessing parameters.

The Modbus Block Write function allows the user to write up to 32 16-Bit parameters in one block transfer.

2 Modbus (RTU)

2.1 Single read/write parameter

2.1.1 Frame Description

Request

Address	Function	Dataset/Parameter	Number of Register	CRC
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Read response

Address	Function	No. Bytes	Parameter Value	CRC
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The whole code is written in hexadecimal digits

The **Address** field contains the slave-address in the Modbus message. Valid slave node addresses are the range of 1-247. A master addresses a slave by placing the slave address in the address field of the message. When the slave returns its response, it places its own address in the response address field to let the master know which slave is responding.

The Address is set in Parameter 1376.

The **Function** Code indicates to the inverter what kind of action to perform. The most important functions for Bonfiglioli Inverter are:

Function 03 (0x03) to read out 16 and 32bit parameter / Block read

Function 06 (0x06) to write single 16bit parameter

Function 16 (0x10) to write single 32bit parameter / Block write

Dataset/Parameter is using the following algorithm:

DataSet/Parameter = parameter number + (data set * 4096)

The Datasets are connected as followed:

EEPROM	RAM
Dataset 0	Dataset 5
Dataset 1	Dataset 6
Dataset 2	Dataset 7
Dataset 3	Dataset 8
Dataset 4	Dataset 9

The actual Values are always stored in Dataset 0 respectively Dataset 5.

Writing to Dataset 0 means that the same value will be written to Dataset 1 to 4. Writing to Dataset 5 means that the same value will be written to dataset 6 to 9.

For example to write dataset 5 of parameter 222 ($222 + 5 \cdot 4096 = 20702$) you have to put "50 DE".

If it is necessary to write parameter **permanently/automatically** to the inverter, it is **indispensable** to write them to dataset 5,6,7,8 or 9. These Datasets are located on the internal RAM which will be **erased** after a reboot/shutdown of the inverter. If you would write continuously to dataset 0 to 4 you would write to the internal EEPROM which is limited to 1000000 writings. The only exception is, when you have to set a parameter manually/once to a value to do adjustments or similar.

For automatic write processes, it is necessary to use dataset 5-9 and you must not use dataset 0-4. Otherwise you will destroy the controller of the AEC and it needs to be replaced.

The inverter is always working in data set 1 respectively 6. To write parameters it is recommended to use data set 5.

No.Bytes For 16bit values there will be 2 Bytes; for 32bit values there will be 4 Bytes

Number of Register: To read out or to write 16bit values the number of register is 1. For 32bit values the number of register is 2.

Parameter Value The value of the read out value or the new value of the parameter.

The **CRC** is the result of a "Redundancy Checking" calculation that is performed on the message contents.

To calculate the **Cyclic Redundancy Check** please check <http://www.simplymodbus.ca/>

2.1.2 Examples

2.1.2.1 Read 16bit Register/Parameter

Example read parameter 213 Active Power (16bit) Dataset 0 of Inverter with Address 1

Request

Address	Function	Dataset/Parameter		Number of Register		CRC	
01	03	00	D5	00	01	95	F2

Response of Inverter with Address 1 with 30kW

Address	Function	No.Bytes	Parameter Value		CRC	
01	03	02	01	2C	B8	09

2.1.2.2 Write 16bit Register/Parameter

Example set parameter 1020 Power reduction reference value (16bit) Dataset **5** of Inverter with Address 1 to 85%

Request

Address	Function	Dataset/Parameter		Parameter Value		CRC	
01	06	53	FC	00	55	98	81

Response

Address	Function	Dataset/Parameter		Parameter Value		CRC	
01	06	53	FC	00	55	98	81

2.1.2.3 Read 32bit Register/Parameter

Example read parameter 1201 "Response Time TDG Undervoltage 2" (32bit) Dataset 0 of Inverter with Address 1

Request

Address	Function	Dataset/Parameter		Number of Register		CRC	
01	03	04	B1	00	02	95	1C

Response of Inverter with Address 1 with 200 ms

Address	Function	No.Bytes	Parameter Value				CRC	
01	03	04	00	00	00	C8	FB	A5

2.1.2.4 Write 32bit Register/Parameter

Example set parameter 1201 "Response Time TDG Undervoltage 2" (32bit) Dataset **5** of Inverter with Address 1 to 25000 ms

Request

Address	Function	Dataset/Parameter		No. of Register		No. of Bytes	Parameter Value			CRC	
01	10	54	B1	00	02	04	00	00	61	A8	DF FA

Response

Address	Function	Dataset/Para.		No. of Register		CRC	
01	10	54	B1	00	02	01	DF

Block Read Function

The Modbus Function 0x03 Read Holding Registers will be extended for the reading of up to 64 inverter parameters (16-Bit) in a single block transfer. This is used to reduce the Bus traffic.

The Modbus Block Read Function will allow the reading of up to 64 Inverter Parameters which correspond to contiguous Modbus Register from Reg.No. 0xF01 to 0xF40 (3841-3904 dec.)

As the register numbers do not directly refer to the inverter parameter numbers, a mapping of the register to inverter parameters is required.

This will be carried out using two Index-Parameter with 32 indices each, each containing an inverter parameter number selected. The indices 1 – 32 of the first index parameter refer to the register numbers 0xF01 – 0xF20 and the indices of the second index parameter refer to the register numbers 0xF21 – 0xF40 respectively.

All parameters corresponding to the Registers defined in the request are read in sequence and the data values then stored in the "Data" field of the Response Frame.

2.1.3 Possible Block Read Parameter

In VTable it is easy to define Index 1-32 of parameters 1282 and 767 (Block Read Parameter). It is possible to set each 16 bit parameter that can be read. Note that only Dataset 0 can be read. String parameter and 32 bit parameter are not possible.

*The Inverter Parameter **P.301** and **P.850** are 32-bit Parameter, but can also be used in block mode. They are converted to 16-Bit.*

The frequency (P.850) value has 4 decimal places and is transferred without the decimal point, i.e. the value is multiplied by 10000 (e.g. a frequency value of 50,0025 Hz is stored as 500025 which corresponds to 0x7A139 in HEX format. In order to convert this value in 16-Bits it will be cut to two decimal places, i.e. the stored value will be divided by 100. Using the example above 500025 (50,0025Hz) will be transferred as 5000 (50,00Hz) thus losing the last two decimal places.

The Active Energy (P.301) has no decimal places. The convention is done by dividing the value by 1000 respectively by changing from kWh to MWh (e.g. a active Energy value of 6189 kWh will be transferred as 6MWh.

In addition to the Parameters the following value can be mapped to the Modbus read mode:

2000 / 2001, "Low and High Word of P301 (Energy positive)"

3000 / 3001, "Low and High Word of P244 (Working hours)"

4000 / 4001, "Low and High Word of P345 (Operating hours)"

5000 / 5001, "Low and High Word of P302 (Energy negative)"

2.1.4 Frame Description

Request

Address	Function	Dataset/Parameter		Number of Register		CRC	
01	03	0F	01	00	04	16	DD

Response

Address	Function	No.Bytes	Data			CRC	
01	03	04	00	...	50	AA	FF

Address and **Function** see above

Dataset/Parameter Startposition to read out the predefined Block read Parameter 1282.

"0F 01" for Index 1 – "0F 20" for Index 32

Number of Register Among of parameter to read out.

No.Bytes depends on the among of the requested parameters (0x02 – 0x40)

Data:

Hi-Byte 1 st .Par.	Lo-Byte 1 st Par.	...	Hi-Byte Nth. Par.	Lo-Byte Nth par.
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N = No. of Registers

2.1.5 Factory Settings

P.1282 Index	Factory Setting	Corresponding Modbus Reg.No
1	213 – Active Power	0xF01
2	222 - DC-Link Voltage	0xF02
3	255 – Heat Sink Temperature	0xF03
4	256 – Inside Temperature	0xF04
5	260 – Current Error	0xF05
6	852 – Power Supply Current	0xF06
7	853 – Power Supply Voltage	0xF07
8	855 – DC Power	0xF08
9	879 – Apparent Power	0xF09
10	1090 – Solar Status	0xF0A
11	850 – Frequency	0xF0B
12	250 – Digital Inputs	0xF0C
13	860 – DC Current	0xF0D
14	301 – Active Energy	0xF0E
15	875 – Apparent Power a	0xF0F
16	876 – Apparent Power b	0xF10
17	877 – Apparent Power c	0xF11
18	869 – Active Power a	0xF12
19	870 – Active Power b	0xF13
20	871 – Active Power c	0xF14
21	863 – Current a	0xF15
22	864 – Current b	0xF16
23	865 – Current c	0xF17

24	866 – Power Supply Voltage a	0xF18
25	867 – Power Supply Voltage b	0xF19
26	868 – Power Supply Voltage c	0xF1A
27	872 – Reactive Power a	0xF1B
28	873 – Reactive Power b	0xF1C
29	874 – Reactive Power c	0xF1D
30	878 – Reactive Power	0xF1E
31	1161- Reference Value Reac-	0xF1F
32	222 - DC-Link Voltage	0xF20

P.767 Index	Factory Setting	Corresponding Modbus Reg.No
1	211 – x	0xF21
2	223 – x	0xF22
3	229 – x	0xF23
4	2000 – x	0xF24
5	2001 – x	0xF25
6	243 – x	0xF26
7	249 – x	0xF27
8	251 – x	0xF28
9	253 – x	0xF29
10	254 – x	0xF2A
11	257x	0xF2B
12	266 – x	0xF2C
13	277 – x	0xF2D
14	222 - x	0xF2E
15	3000 – x	0xF2F
16	3001 – x	0xF30
17	4000 – x	0xF31
18	4001 – x	0xF32
19	291 – x	0xF33
20	292 – x	0xF34
21	293 – x	0xF35
22	294 – x	0xF36
23	295 – x	0xF37
24	296 – x	0xF38
25	297 – x	0xF39
26	851 – x	0xF3A
27	856 – x	0xF3B
28	857 – x	0xF3C
29	858 – x	0xF3D
30	859 – x	0xF3E
31	861 – x	0xF3F
32	862- x	0xF40

2.1.6 Example

2.1.6.1 Read 4 Parameter in one Block transfer starting at Index 1

Request

Address	Function	Dataset/Parameter		Number of Register		CRC	
01	03	0F	01	00	04	16	DD

Response

Address	Function	No.Bytes	Data								CRC	
01	03	08	02	6C	15	85	02	46	01	A1	57	56

With factory settings:

P.213 = 02 6C → 620 → 62,0kW

P.222 = 15 85 → 5509 → 550,9V

P.255 = 02 46 → 582 → 58,2°C

P.256 = 01 A1 → 417 → 41,7°C

2.2 Block Write Function

The Modbus Function 0x16 Write Holding Registers will be extended for the writing of up to 32 inverter parameters (16-Bit) in a single block transfer. This is used to reduce the Bus traffic.

The Modbus Block Write Function will allow the writing of up to 32 Inverter Parameters which correspond to contiguous Modbus Register from Reg.No. 0xF51 to 0xF70 (3921 - 3952 dec.)

As the register numbers do not directly refer to the inverter parameter numbers, a mapping of the register to inverter parameters is required.

This will be carried out using an Index-Parameter with 32 indices, each containing an inverter parameter number selected. The indices 1 – 32 of the index parameter refer to the register numbers 0xF51 – 0xF70 respectively.

All parameters corresponding to the Registers defined in the request are written in sequence and the data values.

2.2.1 Possible Block Write Parameter

In VTable it is easy to define Index 1-32 of parameters 600 (Block Write Parameter). It is possible to set each 16 bit parameter that can be written. Note that also the dataset can be defined the following way:

Value = Parameternumber + DS * 4096

All parameters written with Block Write Mode are not written into the EEPROM. That means that there is no difference between DS0 and DS5, DS1 and DS6 etc. Note that this is only valid for Block Write Mode.

String parameter and 32 bit parameter are not possible.

2.2.2 Frame Description

Request

Address	Function	Parameter		#Register	# Bytes	Data			CRC	
01	10	0F	51	00	04	08	00	...	00	XX XX

Response

Address	Function	Parameter		#Register	CRC		
01	10	0F	51	00	04	XX	XX

Address and **Function** see above

Dataset/Parameter Startposition to read out the predefined Block read Parameter 600.

"0F 51" for Index 1 – "0F 70" for Index 32

Number of Register Among of parameter to read out.

No.Bytes depends on the among of the requested parameters (0x02 – 0x40)

Data:

Hi-Byte 1 st .Par.	Lo-Byte 1 st Par.	...	Hi-Byte Nth. Par.	Lo-Byte Nth par.
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N = No. of Registers

2.2.3 Factory Settings

P.1282 Index	Factory Setting	Corresponding Modbus Reg.No
1 - 32	524 – Fix Percentage Value	0xF51 – 0xF70

2.2.4 Example

2.2.4.1 Read 4 Parameter in one Block transfer starting at Index 1

Request

Address	Function	Parameter		#Register		# Bytes	Data				CRC	
01	10	0F	51	00	02	04	00	01	00	02	XX	XX

With factory settings:

P.524 = 1

P.524 = 2

Response

Address	Function	Parameter		#Register		CRC	
01	10	0F	51	00	02	XX	XX

3 Modbus TCP

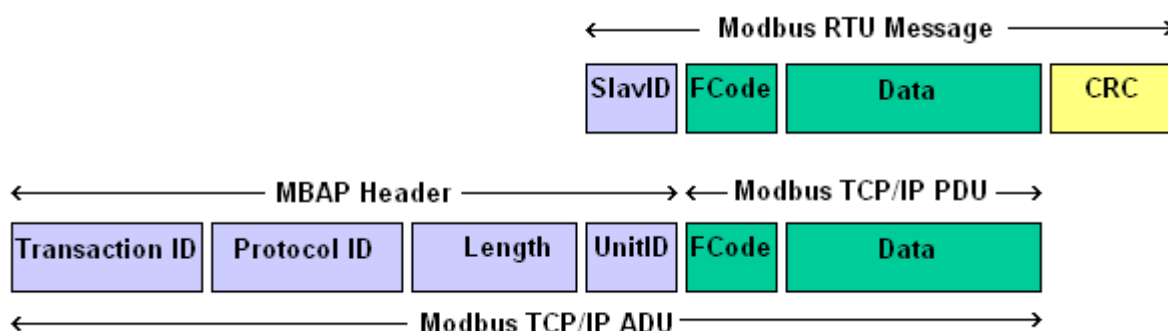
The frequency inverter can be controlled from a PLC or another master device via an Ethernet interface and a serial to Ethernet Modbus gateway, using Modbus/TCP protocol

The MODBUS/TCP communication protocol is a Client/Server related protocol. A Modbus/TCP communication is always initiated by the client (e.g. PLC). The server nodes (inverters) do not communicate with each other.

Modbus/TCP communication is established by Modbus/TCP client to TCP/IP port #502 on side of Modbus/TCP server.

3.1 Message Structure

A Modbus/TCP message is made up of the following fields:



MBAP Modbus Application Header

Field	Length	Description	Client	Server (inverter)
Transaction Identifier	2 Bytes	Identification of a Modbus request/response transaction	Initialized by client	Recopied by the server from received
Protocol Identifier	2 Bytes	0 = Modbus protocol	Initialized by client	Recopied by the server from received
Length	2 Bytes	Number of following bytes (including Unit Identifier)	Initialized by client (request)	Initialized by server (response)
Unit Identifier	1 Byte	Identification of a remote slave connected on a serial line	Initialized by client (request)	Initialized by server (response)

The **Transaction ID** is a counter, so that the master can assign the responses the slaves.

The **Protocol Identifier** is set to 0 for Modbus

The **Length** depends on the frame Designation of the Modbus Function. For Function 0x03;0x06 and 0x65 it is always 0x0006

The **Unit Identifier** is equal to the Slave Address of Modbus RTU

3.2 Example

Request Parameter 222 DC-Link Voltage Dataset 0 2nd Transaction

Transaction ID		Protocol ID		Length		Unit ID	Fun. Code	Dataset/Para.		Number of Register	
00	02	00	00	00	06	01	03	00	DE	00	01

Response P.222 → 0x0063 = 9,9V

Transaction ID		Protocol ID		Length		Unit ID	Fun. Code	No. Bytes	Para. Value	
00	02	00	00	00	05	01	03	02	00	63

4 Inverter Setup

4.1 Protocol

To set the Protocol do the following.

Under "Special Function" → "RS485/RS232" set the Parameter 395 "Protocol" to Value "2 – Modbus-RTU" In case of SyCo this settings is not necessary and must not be set.